9.3 Geometric Series Worksheet — Solutions

Geometric Series

A geometric series has the form

$$a + ar + ar^2 + \cdots + ar^{k-1} + \cdots,$$

where each term is obtained by multiplying the previous term by the same number r.

How to find a and r:

1. $a =$ **the first term** of the series. Plug in the *starting* value of k (which may be $k = 0$, $k = 1$, $k = 2$, …).
2. $r =$ **(second term) ÷ (first term)**. To be sure the series is geometric, also check that the ratio of *any* two consecutive terms, $\dfrac{u_{k+1}}{u_k}$, is the same constant r.
3. If $|r| < 1$, the series converges and its sum is $\dfrac{a}{1-r}$. If $|r| \geq 1$, the series diverges.

---

1. $\displaystyle\sum_{k=0}^{\infty} \frac{7}{3^k} = 7 + \frac{7}{3} + \frac{7}{3^2} + \cdots + \frac{7}{3^k} + \cdots$

Find a. The series starts at $k = 0$, so the first term is

$$a = \frac{7}{3^0} = \frac{7}{1} = 7.$$

Find r. The second term ($k = 1$) is $\dfrac{7}{3}$, so

$$r = \frac{\text{second term}}{\text{first term}} = \frac{7/3}{7} = \frac{1}{3}.$$

Check with the general term: $\dfrac{u_{k+1}}{u_k} = \dfrac{7/3^{k+1}}{7/3^k} = \dfrac{3^k}{3^{k+1}} = \dfrac{1}{3}$ for every k. ✓

Converge or diverge? $|r| = \frac{1}{3} < 1$, so the series converges and

$$\sum_{k=0}^{\infty} \frac{7}{3^k} = \frac{a}{1-r} = \frac{7}{1-\frac{1}{3}} = \frac{7}{\frac{2}{3}} = 7 \cdot \frac{3}{2} = \frac{21}{2}.$$

Answer: $\boxed{\text{converges, sum} = \dfrac{21}{2}}$

---

2. $\displaystyle\sum_{k=0}^{\infty} \frac{(-1)^k\, 6}{5^k} = 6 - \frac{6}{5} + \frac{6}{5^2} - \cdots + \frac{(-1)^k\, 6}{5^k} + \cdots$

Find a. The series starts at $k = 0$:

$$a = \frac{(-1)^0 \cdot 6}{5^0} = \frac{1 \cdot 6}{1} = 6.$$

Find r. The second term ($k = 1$) is $\dfrac{(-1)^1 \cdot 6}{5} = -\dfrac{6}{5}$, so

$$r = \frac{-6/5}{6} = -\frac{1}{5}.$$

Check: $\dfrac{u_{k+1}}{u_k} = \dfrac{(-1)^{k+1}\,6/5^{k+1}}{(-1)^k\,6/5^k} = (-1)\cdot\dfrac{5^k}{5^{k+1}} = -\dfrac{1}{5}$. ✓
(The alternating signs give a *negative* ratio.)

Converge or diverge? $|r| = \frac{1}{5} < 1$, so the series converges and

$$\sum_{k=0}^{\infty} \frac{(-1)^k\,6}{5^k} = \frac{a}{1-r} = \frac{6}{1-\left(-\frac{1}{5}\right)} = \frac{6}{\frac{6}{5}} = 5.$$

Answer: $\boxed{\text{converges, sum} = 5}$

---

3. $\displaystyle\sum_{k=1}^{\infty} 2^{2k}\,3^{1-k} = 4 + \frac{16}{3} + \frac{64}{9} + \cdots + 2^{2k}\,3^{1-k} + \cdots$

Find a. The series starts at $k = 1$:

$$a = 2^{2(1)}\cdot 3^{1-1} = 2^2\cdot 3^0 = 4\cdot 1 = 4.$$

Find r. The second term ($k = 2$) is $2^4\cdot 3^{-1} = \dfrac{16}{3}$, so

$$r = \frac{16/3}{4} = \frac{16}{12} = \frac{4}{3}.$$

Check (concealed form): $2^{2k}\,3^{1-k} = \dfrac{4^k}{3^{k-1}} = 4\cdot\dfrac{4^{k-1}}{3^{k-1}} = 4\left(\dfrac{4}{3}\right)^{k-1}$, which is ar^{k-1} with $a = 4$, $r = \frac{4}{3}$. ✓

Converge or diverge? $|r| = \frac{4}{3} \geq 1$, so the series diverges.

Answer: $\boxed{\text{diverges}}$

---

4. $\displaystyle\sum_{k=1}^{\infty} \frac{5^{k+1}}{8^k} = \frac{25}{8} + \frac{125}{64} + \frac{625}{512} + \cdots + \frac{5^{k+1}}{8^k} + \cdots$

Find a. The series starts at $k = 1$:

$$a = \frac{5^{1+1}}{8^1} = \frac{25}{8}.$$

Find r. The second term ($k = 2$) is $\dfrac{5^3}{8^2} = \dfrac{125}{64}$, so

$$r = \frac{125/64}{25/8} = \frac{125}{64}\cdot\frac{8}{25} = \frac{5}{8}.$$

Check: $\dfrac{u_{k+1}}{u_k} = \dfrac{5^{k+2}/8^{k+1}}{5^{k+1}/8^k} = \dfrac{5^{k+2}}{5^{k+1}}\cdot\dfrac{8^k}{8^{k+1}} = 5\cdot\dfrac{1}{8} = \dfrac{5}{8}$. ✓

Converge or diverge? $|r| = \frac{5}{8} < 1$, so the series converges and

$$\sum_{k=1}^{\infty} \frac{5^{k+1}}{8^k} = \frac{a}{1-r} = \frac{\frac{25}{8}}{1-\frac{5}{8}} = \frac{\frac{25}{8}}{\frac{3}{8}} = \frac{25}{3}.$$

Answer: $\boxed{\text{converges, sum} = \dfrac{25}{3}}$

---

5. $\displaystyle\sum_{k=2}^{\infty} \frac{3^k}{2^{3k-1}} = \frac{9}{32} + \frac{27}{256} + \frac{81}{2048} + \cdots + \frac{3^k}{2^{3k-1}} + \cdots$

Find a. Careful: this series starts at $k = 2$, so the first term is

$$a = \frac{3^2}{2^{3(2)-1}} = \frac{9}{2^5} = \frac{9}{32}.$$

(A common mistake is to use $k = 1$ or $k = 0$ here.)

Find r. The second term ($k = 3$) is $\dfrac{3^3}{2^8} = \dfrac{27}{256}$, so

$$r = \frac{27/256}{9/32} = \frac{27}{256} \cdot \frac{32}{9} = \frac{3}{8}.$$

Check: $\dfrac{u_{k+1}}{u_k} = \dfrac{3^{k+1}/2^{3k+2}}{3^k/2^{3k-1}} = \dfrac{3^{k+1}}{3^k} \cdot \dfrac{2^{3k-1}}{2^{3k+2}} = 3 \cdot \dfrac{1}{2^3} = \dfrac{3}{8}$. ✓

Converge or diverge? $|r| = \frac{3}{8} < 1$, so the series converges and

$$\sum_{k=2}^{\infty} \frac{3^k}{2^{3k-1}} = \frac{a}{1-r} = \frac{\frac{9}{32}}{1-\frac{3}{8}} = \frac{\frac{9}{32}}{\frac{5}{8}} = \frac{9}{32} \cdot \frac{8}{5} = \frac{9}{20}.$$

Answer: $\boxed{\text{converges, sum} = \dfrac{9}{20}}$

---

6. $3 + \dfrac{3}{4} + \dfrac{3}{4^2} + \cdots + \dfrac{3}{4^{k-1}} + \cdots$

Find a. Read the first term directly from the list: $a = 3$.
(In sigma notation the series is $\sum_{k=1}^{\infty} \frac{3}{4^{k-1}}$, and $k = 1$ gives $\frac{3}{4^0} = 3$.)

Find r. The second term is $\dfrac{3}{4}$, so

$$r = \frac{3/4}{3} = \frac{1}{4}.$$

Check: $\dfrac{3}{4^{k-1}} = 3\left(\dfrac{1}{4}\right)^{k-1}$, which is ar^{k-1} with $a = 3$, $r = \frac{1}{4}$. ✓

Converge or diverge? $|r| = \frac{1}{4} < 1$, so the series converges and

$$\frac{a}{1-r} = \frac{3}{1-\frac{1}{4}} = \frac{3}{\frac{3}{4}} = 4.$$

Answer: $\boxed{\text{converges, sum} = 4}$

---

7. $\dfrac{2}{9} - \dfrac{4}{27} + \dfrac{8}{81} - \cdots + \dfrac{(-1)^{k-1}\,2^k}{3^{k+1}} + \cdots$

Find a. Read the first term directly: $a = \dfrac{2}{9}$.

(With the general term starting at $k = 1$: $\frac{(-1)^0\,2^1}{3^2} = \frac{2}{9}$.)

Find r. The second term is $-\dfrac{4}{27}$, so

$$r = \frac{-4/27}{2/9} = -\frac{4}{27} \cdot \frac{9}{2} = -\frac{2}{3}.$$

Check with the third term: $\dfrac{8/81}{-4/27} = \dfrac{8}{81} \cdot \left(-\dfrac{27}{4}\right) = -\dfrac{2}{3}$. ✓

In general, $\dfrac{(-1)^{k-1}\,2^k}{3^{k+1}} = \dfrac{2}{9}\left(-\dfrac{2}{3}\right)^{k-1}$.

Converge or diverge? $|r| = \frac{2}{3} < 1$, so the series converges and

$$\frac{a}{1-r} = \frac{\frac{2}{9}}{1-\left(-\frac{2}{3}\right)} = \frac{\frac{2}{9}}{\frac{5}{3}} = \frac{2}{9} \cdot \frac{3}{5} = \frac{2}{15}.$$

Answer: $\boxed{\text{converges, sum} = \dfrac{2}{15}}$